Kısmi Integrasyon

Eğer f ve g; x'in türevlenebilir fonksiyonları ise,
Çarpım kuralı

$$\frac{d}{dx}\Big(f(x)g(x)\Big) = f'(x)g(x) + f(x)g'(x)$$

$$\int \frac{d}{dx}\Big(f(x)g(x)\Big)\,dx = \int \Big(f'(x)g(x) + f(x)g'(x)\Big)\,dx$$

buradan

$$\int f(x)\,g'(x)\,dx = f(x)g(x) - \int f'(x)\,g(x)\,dx$$

Bu formül diferansiyel formunda yazıldığında daha kolay hatırlanabilir.

$u = f(x)$ ve $v = g(x)$ olsun, bu durumda $du = f'(x)\,dx$

ve $dv = g'(x)\,dx$ olur.

Kısmi Integrasyon Formülü

$$\boxed{\int u\,dv = uv - \int v\,du}$$

Örnek. $\int x\cos x\, dx$ integralini hesaplayınız.

Çözüm.

$$u = x \quad , \quad dv = \cos x\, dx$$

$$du = dx \quad , \quad v = \sin x$$

$$\int x \cos x\, dx = x\sin x - \int \sin x\, dx = x\sin x + \cos x + C$$

Bu örnekte u ve dv için dört türlü seçim mümkündür:

1. $u=1$, $dv = x\cos x\, dx$
2. $u=x$, $dv = \cos x\, dx$
3. $u = x\cos x$, $dv = dx$
4. $u = \cos x$, $dv = x\, dx$

2. seçenek çözümde kullanılmıştır. Geriye kalan üç seçenek nasıl integre edeceğimizi bilmediğimiz integrallere yol açar.

Örneğin 3. seçenek

$$\int (x\cos x - x^2 \sin x)\, dx$$

integralini verir.

Kısmi integrasyonun amacı nasıl hesaplanacağını bilemediğimiz bir $\int u\,dv$ integralinden hesaplayabileceğimiz bir $\int v\,du$ integraline gitmektir.

Örnek. $\int \ln x \, dx$ integralini hesaplayınız.

Çözüm.

$$\int \ln x \, dx = \int \ln x \cdot 1 \, dx$$

$$u = \ln x \quad , \quad dv = dx$$

$$du = \frac{1}{x} dx \quad , \quad v = x$$

$$\int \ln x \, dx = x \ln x - \int x \frac{1}{x} dx = x \ln x - \int dx$$

$$= x \ln x - x + C$$

$\int x^2 e^x \, dx = ?$

$$x^2 = u \,, \qquad e^x dx = d\vartheta$$

$$\Rightarrow 2x\,dx = du \,, \qquad e^x = \vartheta$$

$$\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx$$

$$= x^2 e^x - 2 \int x e^x \, dx$$

$$u = x \,, \qquad e^x dx = d\vartheta$$

$$\Rightarrow du = dx \,, \qquad e^x = \vartheta$$

$$\int x^2 e^x \, dx = x^2 e^x - 2 \int x e^x \, dx = x^2 e^x - 2 \left(x e^x - \int e^x dx \right)$$

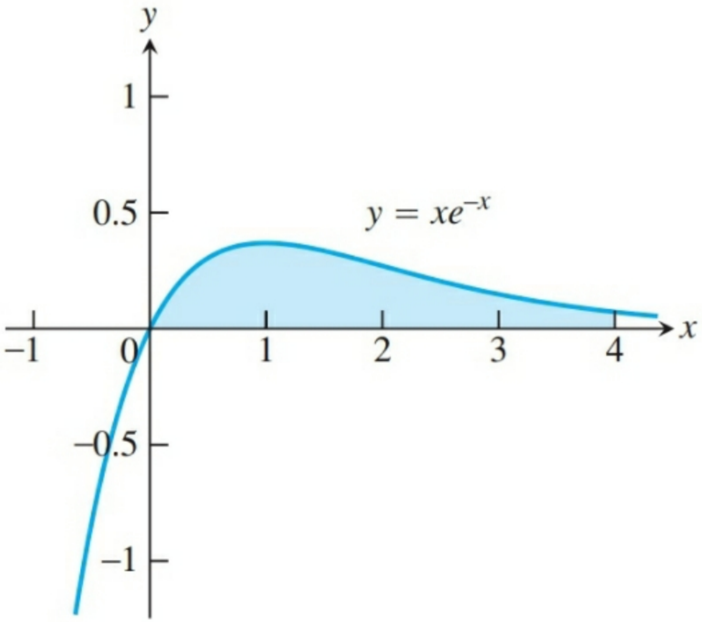
$$= x^2 e^x - 2x e^x + 2 \int e^x + C = x^2 e^x - 2x e^x + 2e^x + C$$

Belirli Integralleri Kısmi Integrasyonla Hesaplamak

Belirli integraller için Kısmi integrasyon formülü:

$$\boxed{\int_a^b f(x)\,g'(x)\,dx = f(x)g(x)\Big|_a^b - \int_a^b f'(x)\,g(x)\,dx}$$

Örnek. $x=0$'dan $x=4$'e kadar $y=xe^{-x}$ eğrisi ve x-ekseni ile sınırlı bölgenin alanını bulunuz.



$$A=\int_0^4 xe^{-x}\,dx$$

$$A = \int_0^4 x e^{-x}\, dx$$

$$u = x, \qquad dv = e^{-x}\, dx$$

$$\Rightarrow \quad du = dx, \qquad v = -e^{-x}$$

$$A = -x e^{-x} \Big|_0^4 - \int_0^4 (-e^{-x})\, dx = -4e^{-4} + \int_0^4 e^{-x}\, dx$$

$$= -4e^{-4} - e^{-x} \Big|_0^4 = -4e^{-4} - e^{-4} - (-e^{0}) = 1 - 5e^{-4}$$

$\int x(\ln x)^3\,dx$ belirsiz integralini bulunuz.

$u=(\ln x)^3 \quad , \qquad x\,dx = dv$

$du = 3(\ln x)^2 \cdot \frac{1}{x}\,dx \quad , \qquad \frac{x^2}{2} = v$

$$\int x(\ln x)^3 dx = \frac{x^2}{2}(\ln x)^3 - \int \frac{x^2}{2}\, 3\,(\ln x)^2 \frac{1}{x}\,dx$$

$$= \frac{x^2}{2}(\ln x)^3 - \frac{3}{2}\int x(\ln x)^2\,dx$$

$u = (\ln x)^2 \quad , \qquad x\,dx = dv$

$du = 2(\ln x)\cdot\frac{1}{x} \quad , \qquad \frac{x^2}{2} = v$

$$I = \frac{x^2}{2}(\ln x)^3 - \frac{3}{2}\left(\frac{x^2}{2}(\ln x)^2 - \int \frac{x^2}{2} \cdot 2\ln x \frac{1}{x}\,dx\right.$$

$$I = \frac{x^2}{2}(\ln x)^3 - \frac{3}{4}x^2(\ln x)^2 - \frac{3}{2}\int x \ln x\,dx$$

$$u = \ln x, \qquad x\,dx = dv$$

$$du = \frac{1}{x}\,dx, \qquad \frac{x^2}{2} = v$$

$$I = \frac{x^2}{2}(\ln x)^3 - \frac{3}{4}x^2(\ln x)^2 - \frac{3}{2}\left(\frac{x^2}{2}\ln x - \int \frac{x^2}{2}\frac{1}{x}\,dx\right)$$

$$I = \frac{1}{2}(\ln x)^3 x^2 - \frac{3}{4}(\ln x)^2 x^2 - \frac{3}{4}(\ln x)x^2 - \frac{3}{8}x^2 + C$$

$\int x^3 e^{x^2} dx$ belirsiz integralini hesaplayınız.

$$x^2 = y \Rightarrow 2x\,dx = dy \Rightarrow x\,dx = \frac{dy}{2}$$

$$\int x^3 e^{x^2} dx = \int x^2 e^{x^2} x\,dx = \int y e^y \frac{1}{2} dy = \frac{1}{2}\int y e^y dy$$

$$y = u \;, \quad e^y dy = dv$$

$$dy = du \;, \quad e^y = v$$

$$I = \frac{1}{2}\left(y e^y - \int e^y dy \right) = \frac{1}{2} y e^y - \frac{1}{2} e^y + C$$

$$I = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + C$$

$\int \frac{\ln(\ln x)}{2x}\,dx$ belirsiz integralini hesaplayınız.

$$\ln x = y \Rightarrow \frac{1}{x}dx = dy$$

$$I = \int \frac{1}{2}\ln y\,dy = \frac{1}{2}\int \ln y\,dy$$

$$\ln y = u\,,\quad dy = dv \Rightarrow \frac{1}{y}dy = du\,,\quad y = v$$

$$I = \frac{1}{2}\left(y\ln y - \int y\,\frac{1}{y}\,dy\right) = \frac{1}{2}y\ln y - y + C$$

$$I = \frac{1}{2}\left(\ln x \cdot \ln(\ln x) - \ln x\right) + C$$

4. $\displaystyle\int_1^e \sin(\ln x)\,dx$ belirli integralini hesaplayınız.

$$\ln x = y \Rightarrow \frac{1}{x}\,dx = dy \quad , \quad dx = x\,dy = e^y\,dy$$

$$\int_1^e \sin(\ln x)\,dx = \int_0^1 \sin(y)\cdot e^y\,dy$$

$$u = e^y \quad , \quad \sin y\,dy = dv$$

$$du = e^y\,dy \quad , \quad -\cos y = v$$

$$I = -e^y\cos y + \int \cos y\; e^y\,dy$$

$u = e^y, \quad \cos y\, dy = dv$

$du = e^y dy, \quad \sin y = v$

$I = -e^y \cos y + \left(e^y \sin y - \int \sin y \; e^y \, dy \right)$

$$\int e^y \sin y \, dy = -e^y \cos y + e^y \sin y - \int e^y \sin y \, dy$$

$$\int_0^1 e^y \sin y \, dy = \frac{1}{2} e^y (\sin y - \cos y) \Big|_0^1$$

$$= \frac{1}{2} e (\sin 1 - \cos 1) + \frac{1}{2} e$$

$\int \frac{1+\cot x}{1+e^{-x}\csc x}\,dx$ belirsiz integralini bulunuz.

$$\frac{1+\cot x}{1+e^{-x}\csc x} = 1+\frac{\cos x}{\sin x} \Big/ 1+\frac{1}{e^x}\cdot\frac{1}{\sin x}$$

$$= \frac{\sin x+\cos x}{\sin x}\cdot\frac{e^x\sin x}{1+e^x\sin x}$$

$$= \frac{e^x\sin x+e^x\cos x}{1+e^x\sin x}$$

$$I = \int \frac{\overbrace{e^x \sin x + e^x \cos x}^{u'}}{\underbrace{1 + e^x \sin x}_{u}} = \int \frac{u'}{u}\, du = \ln|u| + C$$

$$I = \ln|1 + e^x \sin x| + C$$

$\int_0^1 \arctan x \, dx = ?$

$\arctan x = u \quad , \quad dx = dv$

$\frac{1}{1+x^2} dx = du \quad , \quad x = v$

$$I = x \arctan x \Big|_0^1 - \int_0^1 x \frac{1}{1+x^2} dx$$

$$= \arctan 1 - \frac{1}{2} \ln(1+x^2) \Big|_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$\int \sec^3 x \, dx = ?$

$$I = \int \sec x . \sec^2 x dx =$$

$$\downarrow$$

$$u = \sec x , \quad \sec^2 x \; dx = d v$$

$$du = \sec x \tan x , \quad \tan x = v$$

$$I = \sec x \tan x - \int \tan x \sec x \tan x = \sec \tan x - \int \sec x \tan^2 x \, dx$$

$$= \sec x \tan x - \int (\sec^2 x - 1) \sec x \; dx = \int \sec^3 x \, dx - \int \sec x \, dx$$

$$\Rightarrow \int \sec^3 x \; dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \int \sec x \, dx$$

$$\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C$$

$$\# \quad \int e^{ax} \cos bx \, dx = ? \qquad (ab \neq 0)$$

$$u = e^{ax}, \qquad dv = \cos bx \, dx$$

$$du = a e^{ax}, \qquad v = \frac{\sin bx}{b}$$

$$\int e^{ax} \cos bx \, dx = \frac{1}{b} e^{ax} \sin bx - \int \frac{a}{b} e^{ax} \sin bx$$

$$u = e^{ax} \qquad dv = \sin bx \, dx$$

$$\Rightarrow \quad du = a e^{ax}, \qquad v = \frac{-1}{b}\cos bx$$

$$\int e^{ax}\cos bx\, dx = \frac{1}{b} e^{ax}\sin bx - \frac{a}{b}\left[-\frac{1}{b} e^{ax}\cos bx + \int \frac{1}{b}\cos bx \, a e^{ax}\right.$$

$$\Rightarrow$$

$$\int e^{ax}\cos bx\, dx = \frac{e^{ax}\sin bx}{b} + \frac{a e^{ax}\cos bx}{b^2} - \frac{a^2}{b^2}\int e^{ax}\cos bx\, dx$$

$$\left(1 + \frac{a^2}{b^2}\right)\int e^{ax}\cos bx\, dx = e^{ax} \quad \frac{a\cos bx + b\sin bx}{b^2}$$

$$\boxed{\int e^{ax}\cos bx\,dx = e^{ax}\,\frac{a\cos bx + b\sin bx}{a^2+b^2} + C}$$

$$\boxed{\int e^{ax}\sin bx\,dx = e^{ax}\,\frac{a\sin bx - b\cos bx}{a^2+b^2} + C}$$ (ödev)

7.3 İNDİRGEME FORMÜLLERİ

$$\boxed{\int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx , \quad n = 2, 3, \ldots}$$

$$\boxed{\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx , \quad n = 2, 3}$$

drnekl. $\int \sin^n x\, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x\, dx \; , \; n = 2,3,\ldots$

$$\int \sin^n x\, dx = \int \sin^{n-1} x \quad \sin x\, dx$$

$$u = \sin^{n-1} x \, , \qquad \sin x\, dx = dv$$

$$du = (n-1) \sin^{n-2} x \, . \cos x\, dx \, , \qquad v = -\cos x$$

$$I = -\cos x \sin^{n-1} x + \int (n-1) \sin^{n-2} x \cos^2 x\, dx$$

$$I = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x\, dx$$

$$\cos^2 x = 1 - \sin^2 x$$

$$I = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, (1 - \sin^2 x) \, dx$$

$$\int \sin^n x \, dx = -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx$$

$$\Rightarrow \quad n \int \sin^n x \, dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \, dx$$

$$\Rightarrow \quad \int \sin^n x \, dx = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

Örnek 2. $\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1}x\,\sin x + \frac{n-1}{n}\int \cos^{n-2}x\,dx$, $n = 2, 3$

$$\int \cos^n x\,dx = \int \cos^{n-1}x \cdot \cos x\,dx$$

$u = \cos^{n-1}x$, $\cos x\,dx = dv$

$du = (n-1)\cos^{n-2}(-\sin x)\,dx$

$v = \sin x$

$$\int \cos^n x\,dx = \cos^{n-1}x\,\sin x + \int (n-1)\cos^{n-2}x \cdot \sin^2 x\,dx$$

$$\int \cos^n x\,dx = \cos^{n-1}x\,\sin x + (n-1)\int \cos^{n-2}x \cdot (1-\cos^2 x)\,dx$$

$$\int \cos^n x \, dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx - (n-1) \int \cos^n x \, dx$$

$\Rightarrow$

$$n \int \cos^n x \, dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \, dx$$

$$\int \cos^n x \, dx = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

Örnek 3. $\displaystyle\int \frac{dx}{(x^2+a^2)^n} = \frac{1}{2a^2}\frac{x}{(x^2+a^2)^n} + \frac{2n-1}{2na^2}\int \frac{dx}{(x^2+a^2)^n}$; $a>0$, $n=1,2,\dots$

$$u = \frac{1}{(x^2+a^2)^n}, \quad dv = dx \Rightarrow -\frac{2nx}{(x^2+a^2)^{n+1}}\,dx = du, \quad v = x$$

$$\int \frac{dx}{(x^2+a^2)^n} = \frac{x}{(x^2+a^2)^n} + \int \frac{2nx^2}{(x^2+a^2)^{n+1}} \quad \text{olur.} \quad (*)$$

Ayrıca

$$\int \frac{x^2}{(x^2+a^2)^{n+1}}\,dx = \int \frac{x^2+a^2-a^2}{(x^2+n^2)^{n+1}}\,dx = \int \frac{dx}{(x^2+a^2)^n} - a^2\underbrace{\int \frac{dx}{(x^2+a^2)^{n+1}}}_{(**)}$$

(*) ve (**)'ı beraber kullanırsak;

$$\int \frac{dx}{(x^2+a^2)^n} = \frac{x}{(x^2+a^2)^n} + 2n \int \frac{dx}{(x^2+a^2)^n} - 2na^2 \int \frac{dx}{(x^2+a^2)^{n+1}}$$

7.4 TRIGONOMETRIK INTEGRALLER

Hatırlatma :

$$\sin^2 x = 1 - \cos^2 x, \qquad \cos^2 x = 1 - \sin^2 x$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}$$

$* \int \sin^m x \cos^n x \, dx$ biçiminde integraller

m ve n'nin tek veya çift olmasına göre uygun değişken dönüşümlerini üç duruma ayırabiliriz.

Durum 1. Eğer m tek ise $m = 2k+1$ olarak yazarız:

$$\int \sin^m x \cos^n x \, dx = \int \sin^{2k+1} x \cos^n x \, dx = \int \cos^n x \cdot \sin^{2k} x \cdot \sin x \, dx$$

$$= \int u^n \cdot (1-u^2)^k \, (-du)$$

$u = \cos x$

Durum 2. Eğer n tek ve m çift ise $n = 2k+1$ şeklinde yazarız.

$$\int \sin^m x \cos^n x \, dx = \int \cos^{2k+1} x \sin^m x \, dx = \int \cos^{2k} x \sin^m x \cdot \cos x dx$$

$u = \sin x$

$$= \int (1-u^2)^k \cdot u^m \, du$$

Durum 3. m ve n'nin her ikisi de çift ise

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \cos^2 x = \frac{1+\cos 2x}{2}$$

dönüşümlerini kullanırız.

Örnek. $\int \sin^3 x \cos^2 x \, dx = ?$

$u = \cos x \Rightarrow du = -\sin x \, dx$

$$I = \int \sin^2 x \cos^2 x \sin x \, dx = \int (1-\cos^2 x) \cos^2 x \cos x \, dx$$

$$= \int (1-u^2)\, u^2 \, (-du)$$

$$= \int (u^4 - u^2) \, du$$

$$= \frac{u^5}{5} - \frac{u^3}{3} + C$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

Örnek. $\int \cos^5 x \, dx = ?$

$$I = \int \cos^4 x \; \cos x \; dx = \int (1-\sin^2 x)^2 \cos x \, dx = \int (1-u^2)^2 \, du$$

$$u = \sin x$$

$$du = \cos x \, dx$$

$$= \int (1 - 2u^2 + u^4) \, du$$

$$= u - \frac{2}{3} u^3 + \frac{1}{5} u^5 + C$$

$$= \sin x - \frac{2}{3} \sin^3 x + \frac{1}{5} \sin^5 x + C$$

Örnek. $\int \sin^2 x \cos^4 x \, dx = ?$

$$I = \int \frac{1-\cos 2x}{2} \cdot \left(\frac{1+\cos 2x}{2} \right)^2 dx$$

$$= \frac{1}{8} \int (1-\cos 2x) . (1 + 2\cos 2x + \cos^2 2x)$$

$$= \frac{1}{8} \int \left(1 + \cos 2x - \cos^2 2x - \cos^3 2x \right) dx$$

$$= \frac{1}{8} \left(x + \frac{1}{2} \sin 2x - \int \left(\cos^2 2x + \cos^3 2x \right) dx \right.$$

$\cos^2 2x$ terimi için

$$\int \cos^2 2x \, dx = \frac{1}{2}\int (1+\cos 4x)\, dx = \frac{1}{2}\left(x+\frac{1}{4}\sin 4x\right)+C$$

$\cos^3 2x$ terimi için

$$\int \cos^3 2x \, dx = \int (1-\sin^2 2x)\cos 2x \, dx = \frac{1}{2}\int (1-u^2)\, du$$

$u = \sin 2x$

$$= \frac{1}{2}\left(\sin 2x - \frac{1}{3}\sin^3 2x\right)+C$$

$$I = \frac{1}{16}\left(x - \frac{1}{4}\sin 4x + \frac{1}{3}\sin^3 2x\right) + C$$

örnek. $\int \sin^4 x \cos^7 x \, dx = ?$

$$\int \sin^4 x \cos^7 x \, dx = \int \sin^4 x \cdot \cos^6 x \cdot \cos x \, dx$$

$$u = \sin x \quad \Rightarrow \quad du = \cos x \, dx$$

$$I = \int u^4 (1-u^2)^3 \, du = \int u^4 (1 - 3u^2 + 3u^4 - u^6) \, du$$

$$= \int (u^4 - 3u^6 + 3u^8 - u^{10}) \, du$$

$$= \frac{1}{5} u^5 - \frac{1}{7} 3 u^7 + \frac{3}{9} u^9 - \frac{1}{11} u^{11} + C$$

$$= \frac{1}{5} \sin^5 x - \frac{3}{7} \sin^7 x + \frac{1}{3} \sin^9 x - \frac{1}{11} \sin^{11} x + C$$

Örnek. $\displaystyle\int \frac{\sin^5 x}{\sqrt{\cos x}}\,dx = ?$

$$\int \sin^5 x \, \frac{1}{\sqrt{\cos x}}\,dx = \int \frac{1}{\sqrt{\cos x}} \cdot (1-\cos^2 x)^2 \cdot \sin x\,dx$$

$u = \cos x \quad \Rightarrow \quad du = -\sin x\,dx$

$$= \int \frac{1}{\sqrt{u}} (1-u^2)^2 (-du) = -\int \frac{u^4 - 2u^2 + 1}{\sqrt{u}}\,du$$

$$= -\int \left(u^{7/2} - 2u^{3/2} + u^{-1/2}\right) du = -\frac{2}{9}u^{9/2} + \frac{4}{5}u^{5/2} - 2u^{1/2} + C$$

$$= -\frac{2}{9}(\cos x)^{9/2} + \frac{4}{5}(\cos x)^{5/2} - 2(\cos x)^{1/2} + C$$

NOT.

$\tan^2 x = \sec^2 x - 1$ $\sec^2 x = \tan^2 x + 1$	$(\sec x)' = \sec x \tan x$ $(\tan x)' = \sec^2 x$

$\int \tan^m x \sec^n x \, dx$ biçimindeki integraller :

Örnek. $\int \sqrt{\tan x} \sec^4 x \, dx = ?$

$$I = \int \sqrt{\tan x} \sec^2 x \sec^2 x \, dx = \int \sqrt{\tan x} (\tan^2 x + 1) \sec^2 x \, dx$$

$\tan x = u \Rightarrow \sec^2 x \, dx = du$ $\qquad I = \int \sqrt{u} (u^2 + 1) \, du$

$$= \frac{2}{7} u^{7/2} + \frac{2}{3} u^{3/2} + C = \frac{2}{7} (\tan x)^{7/2} + \frac{2}{3} (\tan x)^{3/2} + C$$

drnek. $\int \tan^3 x \sec^5 x \, dx = ?$

$$I = \int \tan^2 x \sec^4 x \sec x \tan x \, dx = \int (\sec^2 x - 1) \sec^4 x (\sec x \tan x) dx$$

$$= \int (\sec^6 x - \sec^4 x) \sec x \tan x \, dx$$

$u = \sec x \Rightarrow du = \sec x \tan x$

$$= \int (u^6 - u^4) \, du = \frac{1}{7} u^7 - \frac{1}{5} u^5 + C$$

$$= \frac{1}{7} \sec^7 x - \frac{1}{5} \sec^5 x + C$$

örnek. $\int \tan^2 x \sec^3 x \, dx = ?$

$$\int \tan^2 x \sec^3 x \, dx = \int (\sec^2 x - 1) \sec^3 x \, dx$$

$$= \underbrace{\int \sec^5 x \, dx}_{I_1} - \underbrace{\int \sec^3 x \, dx}_{I_2} \qquad (*)$$

$$I_1 = \int \sec^5 x \, dx = \int \sec^3 x \sec^2 x \, dx$$

$$u = \sec^3 x \, , \qquad \sec^2 x \, dx = dv$$

$$du = 3\sec^3 x \tan x \, , \qquad v = \tan x$$

$$I_1 = \sec^3 x \tan x - \int 3 \tan^2 x \sec^3 x \, dx$$

$$I_1 = \sec^3 x \tan x - 3\int \tan^2 x \sec^3 x \, dx \qquad (**)$$

(**)'ı (*)' da yazarsak

$$\int \tan^2 x \sec^3 x \, dx = \sec^3 x \tan x - 3\int \tan^2 x \sec^3 x \, dx - \int \sec^3 x \, dx$$

$$\Rightarrow \int \tan^2 x \sec^3 x \, dx = \frac{1}{4}\sec^3 x \tan x - \frac{1}{4}\int \sec^3 x \, dx$$

$$= \frac{1}{4}\sec^3 x \tan x - \frac{1}{8}\sec x \tan x - \frac{1}{8}\ln|\sec x + \tan x| + C$$

NOT. $\int \cot^m x \csc^n x \, dx$ biçimindeki integraller için

aşağıdaki eşitlikler kullanılır.

$$\cot^2 x = \csc^2 x - 1$$

$$\csc^2 x = \cot^2 x + 1$$

Örnek. $\int \cot^3 x \csc^3 x \, dx$

$$= \int \cot^2 x \cdot \csc^2 x \, (\csc x \cot x) \, dx \underset{u = \csc x}{=} \int (u^2 - 1) \, u^2 \, (-du)$$

$$= -\frac{u^5}{5} + \frac{u^3}{3} + C = -\frac{1}{5} \csc^5 x + \frac{1}{3} \csc^3 x + C$$

NOT. $\int \sin ax \cos bx \, dx$, $\int \cos ax \cos bx \, dx$, $\int \sin ax \sin bx \, dx$

biçimindeki integraller için aşağıdaki eşitlikler kullanılır:

$$\sin ax \cos bx = \frac{1}{2}\left[\sin(a+b)x + \sin(a-b)x\right]$$

$$\cos ax \cos bx = \frac{1}{2}\left[\cos(a+b)x + \cos(a-b)x\right]$$

$$\sin ax \sin bx = \frac{1}{2}\left[\cos(a-b)x - \cos(a+b)x\right]$$

drnek.

$$\int \sin 2x \cos 5x \, dx = ?$$

$$= \frac{1}{2} \int \left[\sin (2+5)x + \sin (2-5)x \right] dx$$

$$= \frac{1}{2} \int \sin 7x \, dx - \frac{1}{2} \int \sin 3x \, dx$$

$$= -\frac{1}{14} \cos 7x + \frac{1}{6} \cos 3x + C$$